

Train a Regression Model

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can fit a line from a declared training candidate set.

I can separate training loss from validation comparison.

I can interpret slope and intercept with units.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.



LEARNING TARGET Training points $(x,y)=(1,3),(2,5),(3,7)$. Compare $y=x+1$ and $y=2x+1$ by training MAE; select one, then predict held-out $x=4$, actual=10. Explain what was fitted.

Builder name and experiment date:

Readiness for the connected explanation

How is MSE calculated?

This week's end goal

By the end, I can explain train a regression model using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use train a regression model to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A line predicts y as $m \cdot x + b$. In this classroom search we compare a small declared set of slopes and intercepts; this is not a full least-squares solver over all possible lines. Use training error to choose a candidate, then freeze it before validation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines fitting a line.

3. See how it works

ML MECHANISM

Input → **Model** → **Evidence**

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

Run a starter regression and explain slope



ML

MODEL

A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.



OUT

EVIDENCE

and MAE.

QUESTION Run a starter regression and explain slope, prediction, residual, and MAE.

MODEL IDEA A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Model predicts $\text{height} = 4 + 2 \times \text{weeks}$. Slope 2 means two predicted centimeters per additional week in this data range.

Draw a fresh data $\rightarrow$ model $\rightarrow$ prediction $\rightarrow$ evaluation trace:

A closer explanation

Training pairs are (0, 1), (1, 3), (2, 5). Candidate A has $m = 1, b = 1$, B $m = 2, b = 0$, C $m = 2, b = 1$. Predictions are A [1, 2, 3], B [0, 2, 4], C [1, 3, 5]. Training squared-error sums are 5, 3, 0 respectively, so choose C. Ties would use the earlier listed candidate.

```
training = [(0, 1), (1, 3), (2, 5)]
candidates = [(1, 1), (2, 0), (2, 1)]

def sse(line):
    m, b = line
    return sum((y - (m * x + b)) ** 2 for x, y in training)

chosen = min(candidates, key=sse)
print([sse(line) for line in candidates])
print(chosen)

# [5, 3, 0], then (2, 1); first candidate wins ties.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Connect the picture and the explanation

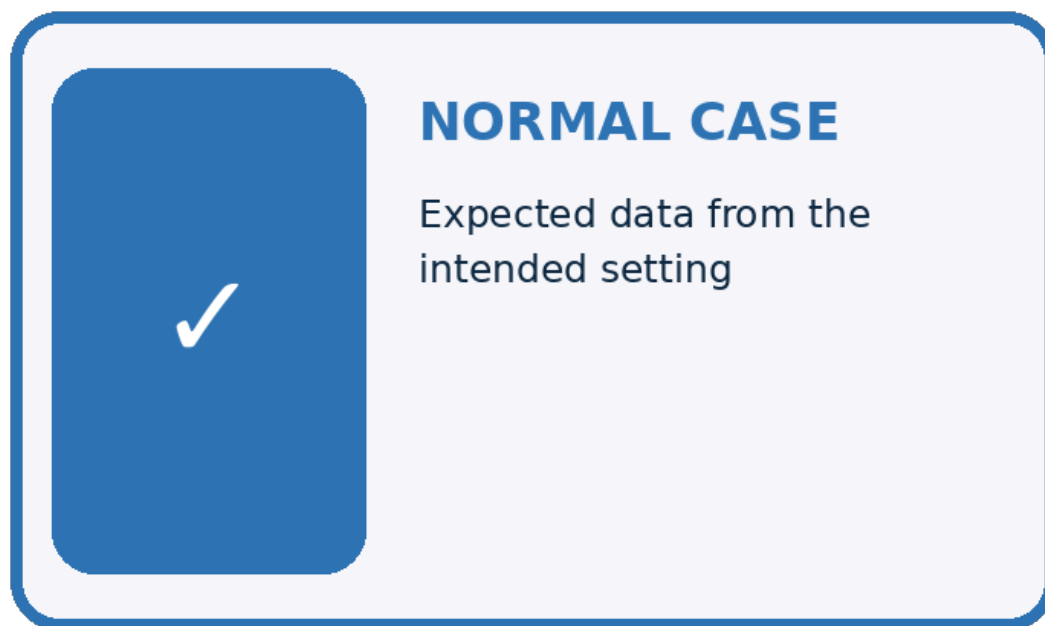
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





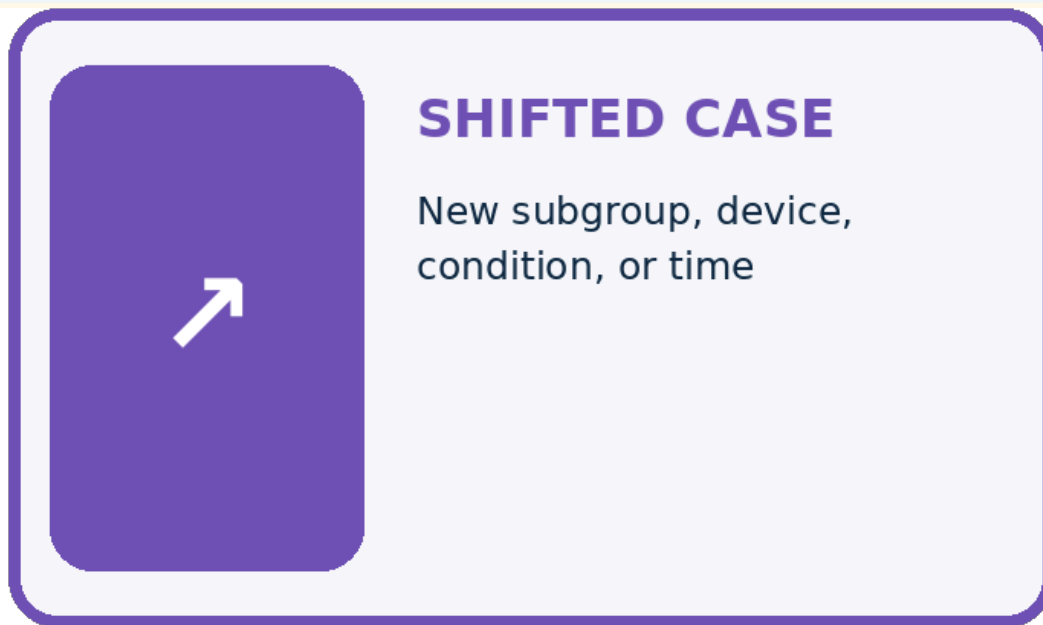
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Evaluation observation becomes 8 instead of 6.

Prediction: Absolute evaluation error stays 1 cm, but residual changes from -1 to +1.

Reason: Prediction₇ remains frozen; changed observation₈ reverses the error direction.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

COEFFICIENT

learned relationship weight

2

INTERCEPT

prediction when inputs are zero

3

SLOPE

predicted change per input unit

4

GENERALIZATION

performance on unseen cases

Term · Precise meaning · My ML example

Term: coefficient

Precise meaning: learned relationship weight

My ML example: _____

Term: intercept

Precise meaning: prediction when inputs are zero

My ML example: _____

Term: slope

Precise meaning: predicted change per input unit

My ML example: _____

Term: generalization

Precise meaning: performance on unseen cases

My ML example: _____

MISCONCEPTION TO REPAIR A positive slope proves one variable causes the other.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Slope

Meaning: Change in predicted target per unit change in x

Word: Intercept

Meaning: Predicted target when x is zero

Word: Candidate set

Meaning: The declared model choices searched

Word: Fitting

Meaning: Selecting model values using training evidence

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 22 • WORKED EVIDENCE

- 1 Model predicts height = $4 + 2 \times \text{weeks}$.
- 2 Slope 2 means two predicted centimeters per additional week in this data range.
- 3 At week 5, prediction is 14 cm; if actual is 16, residual is +2 cm.
- 4 Evaluate unseen plants and compare test MAE with the median baseline.

STEP 1 Model predicts height = $4 + 2 \times \text{weeks}$.

STEP 2 Slope 2 means two predicted centimeters per additional week in this data range.

STEP 3 At week 5, prediction is 14 cm; if actual is 16, residual is +2 cm.

STEP 4 Evaluate unseen plants and compare test MAE with the median baseline.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Fitting a line at a glance

Part · Evidence or action · Interpretation

Part: Line A

Evidence or action: Slope 1 intercept 1

Interpretation: Training SSE 5

Part: Line B

Evidence or action: Slope 2 intercept 0

Interpretation: Training SSE 3

Part: Line C

Evidence or action: Slope 2 intercept 1

Interpretation: Training SSE 0

For separate validation pairs (3, 6), (4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

Predict at x 3 and x 4 and calculate MAE/MSE. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train a Regression Model

Fit a toy growth line from training pairs, then leave evaluation labels out of fitting.

The facts we will use

Training(day, height):(0, 1), (1, 3), (2, 5). Candidates $y=2x+1$ and $y=x+1$. Evaluation pair:(3, 6).

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Train a Regression Model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Calculate training fit

$2x+1$ predicts 1, 3, 5 with SSE0; $x+1$ predicts 1, 2, 3 with SSE5.



2. Select line

Choose $2x+1$ under minimum training SSE.



3. Evaluate once

At day3 predict 7 versus observed 6: absolute error 1 cm.

CHANGE ONE THING

Evaluation observation becomes 8 instead of 6.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Calculate training fit

What happens and why: $2x+1$ predicts 1, 3, 5 with SSE0; $x+1$ predicts 1, 2, 3 with SSE5.

Step: 2. Select line

What happens and why: Choose $2x+1$ under minimum training SSE.

Step: 3. Evaluate once

What happens and why: At day3 predict7 versus observed6: absolute error1 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: A positive slope proves one variable causes the other. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE A positive slope proves one variable causes the other.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Record candidates, training rows, tie policy and chosen parameters. Do not add a candidate after seeing validation and report it as if only the original search happened. A zero training loss can coexist with validation mistakes.

Why is an exact training line not proof of correct predictions at x 100?

8. Practise together

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If x measures days and y measures cm, slope 2 means two predicted cm per day and intercept 1 means one predicted cm at day zero. Predicting far outside the observed x range is extrapolation and may fail even when the training fit is exact.

Practise with less help: Score validation

Use workbook W2 for the connected example “Fitting a line”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Fitting a line. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

May day3’s evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Regression Run: trace code, interpret coefficients with units, predict unseen values, and compare MAE with baseline.

1. Write the hypothesis and expected evidence before running.
2. Regression Run: trace code, interpret coefficients with units, predict unseen values, and compare MAE with baseline.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

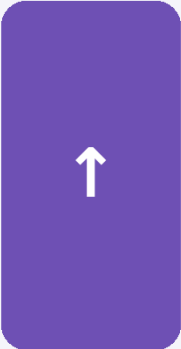
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: coefficient, intercept, slope, generalization.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A positive slope proves one variable causes the other.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Add the regression experiment to the project appendix only as a separate, clearly limited model.

THIS WEEK'S CHECKPOINT Add the regression experiment to the project appendix only as a separate, clearly limited model.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Run a starter regression and explain slope, prediction, residual, and MAE.

REQUIRED PROOF Run a starter regression and explain slope, prediction, residual, and MAE.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Hold Out Test Data.

13. Study a complete case

Week 22 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Supplied fitted rule: $\text{height} = 2 \times \text{age} + 1$. Held-out observations are (3,8),(4,9).

The question

Predict and evaluate; distinguish inference.

Worked explanation

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Apply the fitted equation, then evaluate

Evidence or stage · Value or result

Evidence or stage: Rule

Value or result: $\text{height} = 2 \times \text{age} + 1$

Evidence or stage: Age 3

Value or result: Predict 7, actual 8, absolute error 1

Evidence or stage: Age 4

Value or result: Predict 9, actual 9, absolute error 0

Evidence or stage: MAE

Value or result: 0.5 cm

Supplied fitted rule: $\text{height} = 2 \times \text{age} + 1$. Held-out observations are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

14. Try a new case independently

Independent case: Apply independently

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10),(4, 15) with MAE.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 22 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A regression model has validation MAE 2.4 cm; a training-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add the regression experiment to the project appendix only as a separate, clearly limited model.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Training points $(x,y)=(1,3),(2,5),(3,7)$. Compare $y=x+1$ and $y=2x+1$ by training MAE; select one, then predict held-out $x=4$, actual=10. Explain what was fitted.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Fit and test a straight-line regression model

Fit and test a straight-line regression model

Estimate slope and intercept from training pairs using least squares. Ages are weeks and heights are cm. Test targets are excluded from fitting; training ages must include distinct values.

Math bridge: fitting the line

A straight line predicts $y = \text{slope} \times x + \text{intercept}$. Slope is the change in predicted height for one extra week; intercept is the prediction at $x=0$, which may lie outside the measured ages. You need means, subtraction, multiplication and squared numbers for this example; calculus is not required.

x (weeks) · y (cm) · $x - \text{mean } x$ · $y - \text{mean } y$ · Product · Squared x difference

x (weeks): 1

y (cm): 3

$x - \text{mean } x$: -1

$y - \text{mean } y$: -2

Product: 2

Squared x difference: 1

x (weeks): 2

y (cm): 5

x – mean x: 0

y – mean y: 0

Product: 0

Squared x difference: 0

x (weeks): 3

y (cm): 7

x – mean x: 1

y – mean y: 2

Product: 2

Squared x difference: 1

Mean $x=2$ and mean $y=5$. Add the product column: 4. Add squared x differences: 2. Slope= $4/2=2$ cm per week. Intercept= $5-2\times 2=1$ cm. This least-squares formula chooses the line minimising squared training errors when the x values are not all identical. Compare its held-out error with a baseline; a good training fit alone is insufficient.

```
"""Fit a line to training pairs and evaluate a separate example."""
train = [(1, 3), (2, 5), (3, 7)]

if not train:
    raise ValueError("Training examples are required")

n = len(train)

x_mean = sum(x for x, y in train) / n
y_mean = sum(y for x, y in train) / n

num = sum((x - x_mean) * (y - y_mean) for x, y in train)
den = sum((x - x_mean) ** 2 for x, y in train)

if den == 0:
    raise ValueError("Distinct training ages are required")
```

```
slope = num / den

intercept = y_mean - slope * x_mean

test = [(4, 10)]

if not test:

    raise ValueError("Test examples are required for MAE")

preds = [slope * x + intercept for x, y in test]

mae = sum(abs(y - pred) for pred, (x, y) in zip(preds, test)) / len(test)

print(slope, intercept, preds, mae)
```

Expected output: 2.0 1.0 [9.0] 1.0

The fitted line is $\text{height} = 2 \times \text{age} + 1$. At four weeks it predicts 9 cm; the observed height of 10 cm gives an absolute error and one-case MAE of 1 cm. Fitting a line and applying a supplied line are different activities. One held-out case is very limited evidence.

Independent code change

Change the held-out height from 10 to 8 without changing the training data. Predict the fit, prediction and MAE.

My prediction and evidence

Write or draw your reasoning and evidence.
